

Windows Security Logs Investigation

I started by inspecting the windows-security.log file **first**. Splunk was reading each line as a separate event, so custom line breaking had to be configured using the following regex:

```
(\r\n|\n)(?=d{2}/d{2}/d{4} \d{2}:\d{2}:\d{2} (?AM|PM))
```

The source type was initially set as **uncategorized** so that the raw contents of the events could be reviewed and validated before parsing.

Processes Commonly Associated with T1003.003

The following processes are commonly observed during NTDS.dit extraction attacks:

- ntdsutil.exe
- vssadmin.exe
- wmic.exe
- powershell.exe
- cmd.exe
- diskshadow.exe

Event Code Review

To identify available Windows Security Event Codes, the following search was used:

```
source="windows-security.log" sourcetype="Win_Sec_Log"
| dedup EventCode
| table EventCode
```

A total of 44 unique Event Codes were identified.

4672	4902	4755
	4826	4732
4624	4616	4728
4634	1100	4756
4768	5058	4741
4769	5061	4722
4648	4742	4720
4904	4739	4907
4905	4713	4718
4799	4731	4717
4776	4735	4798
5033	4727	5059
5024	4738	4625
4688	4737	4724
4608	4754	
4781		

New Search

```
source="windows-security.log" sourcetype="Win_Sec_Log"
| dedup EventCode
| table EventCode
```

✓ 44 events (before 6/23/26 6:38:04.000 PM)

No Event Sampling ▾

Job ▾

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⬇

Events

Patterns

Statistics (44)

Visualization

Show: 20 Per Page ▾

✍ Format ▾

☒ Preview: On

EventCode ⬇

4672

4624

4634

4768

4769

4648

4904

4905

4799

4776

5033

5024

4688

4608

Primary Event Codes for T1003.003

Event Code	Description
4663	Object Access (File Read)
4656	Handle Request to Object
4688	Process Creation
5145	Network Share Access
5140	Network Share Object Accessed

Secondary Event Codes

Event Code	Description
8222	Shadow Copy Created
4673	Sensitive Privilege Use
4698	Scheduled Task Created

4657 Registry Value Modified

Findings

The following event codes commonly associated with T1003.003 were searched for:

- 4663 → 0 events
- 4656 → 0 events
- 5145 → 0 events
- 5140 → 0 events
- 8222 → 0 events

No evidence related to NTDS.dit access, shadow copy creation, or network-based NTDS file transfer was found.

Event Code 4688 Analysis

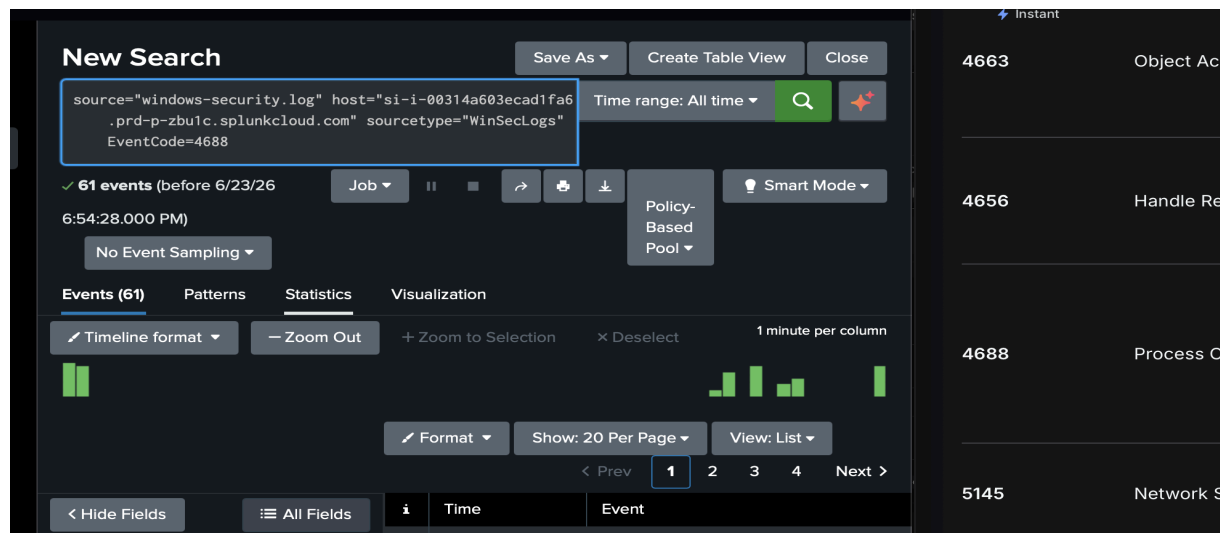
Event Code 4688 had the highest number of potentially relevant events.

Total Events: **61**

Search used:

```
source="windows-security.log" sourcetype="Win_Sec_Log" EventCode=4688  
| rex field=_raw "New Process Name:\s+(?<ProcessName>[^\r\n]+)"
```

```
| table _time, ProcessName
```



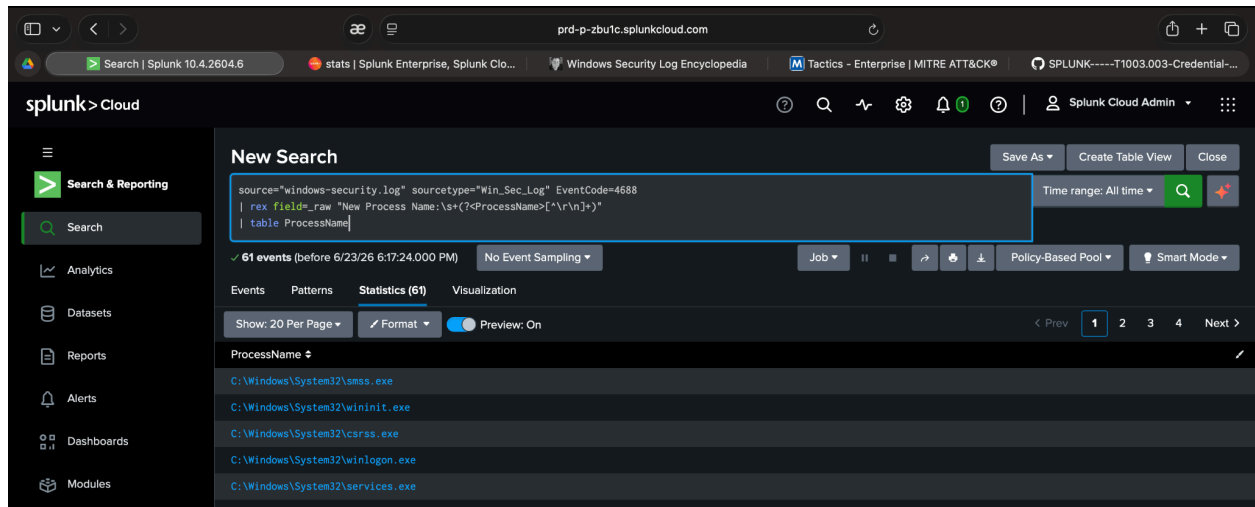
The message field was not parsed correctly by default, so the process names were extracted manually using a regular expression.

i	Time	Event
>	6/19/26 12:52:48.000 PM	10/08/2020 12:52:48 PM LogName=Security SourceName=Microsoft Windows security auditing. EventCode=4688 EventType=0 Type=Information ComputerName=win-dc-8537412.attackrange.local TaskCategory=Process Creation OpCode=Info RecordNumber=228366 Keywords=Audit Success Message=A new process has been created. Creator Subject: Security ID: NT AUTHORITY\SYSTEM Account Name: - Account Domain: - Logon ID: 0x3E7 Target Subject: Security ID: NULL SID Account Name: - Account Domain: - Logon ID: 0x0 Process Information: New Process ID: 0x284 New Process Name: C:\Windows\System32\csrss.exe Token Elevation Type: %%1936 Mandatory Label: Mandatory Label\System Mandatory Level Creator Process ID: 0x27c Creator Process Name: C:\Windows\System32\smss.exe Process Command Line: Token Elevation Type indicates the type of token that was assigned to the new process in accordance with User Account Control policy. Type 1 is a full token with no privileges removed or groups disabled. A full token is only used if User Account Control is disabled or if the process is running with administrative privileges.

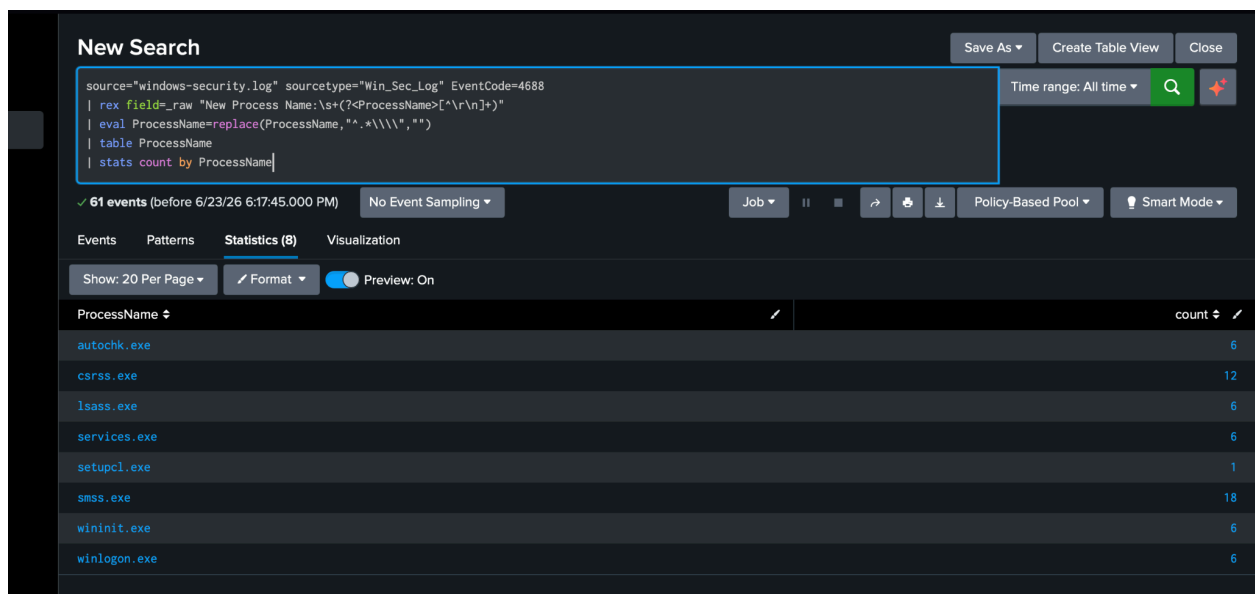
To count all created processes:

```
source="windows-security.log" sourcetype="Win_Sec_Log" EventCode=4688
| rex field=_raw "New Process Name:\s+(?<ProcessName>[^\r\n]+)"
| eval ProcessName=replace(ProcessName,"^.*\\\\", "")
| stats count by ProcessName
```

After reviewing all 61 process creation events, no executions of ntdsutil.exe, vssadmin.exe, diskshadow.exe, wbadmin.exe, or other NTDS-related tools were identified.



```
source="windows-security.log" sourcetype="Win_Sec_Log" EventCode=4688
| rex field=_raw "New Process Name:\s+(?<ProcessName>[^\r\n]+)"
| eval ProcessName=replace(ProcessName,"^.*\\\\","")
| table ProcessName
| stats count by ProcessName
```



Process

Normal role

Malicious?

smss.exe

Session Manager - starts user sessions

No - always runs

csrss.exe	Client/Server Runtime - handles Windows GUI	No - always runs
winlogon.exe	Handles logins	No - always runs
services.exe	Service Control Manager	No - always runs
lsass.exe	Local Security Authority - handles authentication	No - always runs
autochk.exe	Disk checker on boot	No - runs at startup
wininit.exe	Windows initialization	No - runs at startup
setupcl.exe	Windows setup client	No - runs at startup

The observed events appeared normal and did not show suspicious privilege usage related to NTDS.dit extraction.

Conclusion

As a result of the investigation, no evidence of a T1003.003 attack was found within the Windows Security logs.

Out of the 44 unique Event Codes identified, only Event Codes 4688 (Process Creation) and 4672 (Special Privileges Assigned) were potentially relevant to the attack technique. However, detailed analysis showed that all observed processes were legitimate Windows activity and no NTDS-related tools were executed.

Additionally, no events were found for Event Codes 4663, 4656, 5140, 5145, or 8222, which are some of the strongest indicators of NTDS.dit access, shadow copy creation, or NTDS file exfiltration.

Based on the available evidence, no patterns consistent with a T1003.003 NTDS credential dumping attack were identified in the Windows Security logs.